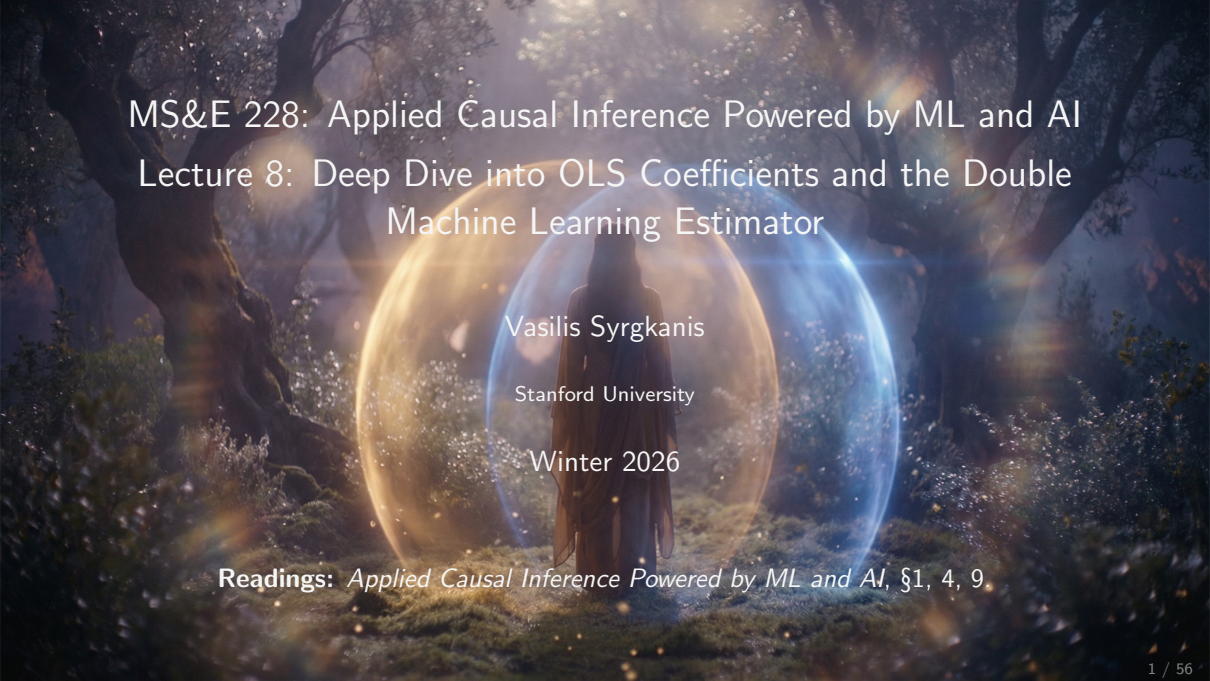

# MS&E 228: Applied Causal Inference Powered by ML and AI

## Lecture 8: Deep Dive into OLS Coefficients and the Double Machine Learning Estimator

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**Readings:** *Applied Causal Inference Powered by ML and AI*, §1, 4, 9.

# Goals for Today

1. **Develop deeper intuition for OLS coefficients:** What does a coefficient in a multivariate regression really mean? When is it uniquely identified, and what drives its variance?
2. **Discover another "insensitive" formula:** Can we simplify inference for OLS coefficients by ignoring nuisance estimation, similar to the DR estimator?
3. **Move beyond full linearity:** How can we allow flexible ML models for covariates while still getting valid causal inference on treatment effects?
4. **Build a practical algorithm for continuous treatments:** What estimator can we use in practice, and what does it estimate when treatment effects are heterogeneous?

# Part I: Understanding OLS Coefficients

The Frisch-Waugh-Lovell Theorem

# Where We Left Off

Last lecture, under full linearity $\mathbb{E}[Y(d)|X] = \theta d + \beta' X$, all causal estimands reduce to estimating $\theta$.

We posed four key questions:

1. How can we interpret $\theta$ intuitively?
2. When is $\theta$ uniquely identified in the best linear predictor?
3. What drives high variance in $\hat{\theta}$?
4. What is the analogue of "overlap" under linearity?

Today we answer all four questions through two marvelous theorems!

# The Frisch-Waugh-Lovell Theorem

**Theorem (Frisch-Waugh-Lovell)**

The coefficient $\hat{\theta}$ on $D$ in $\text{OLS}(Y \sim D, X)$ is identical to the coefficient from:

**Step 1:** Regress $Y$ on $X$ and compute residuals—the "surprise" in $Y$:

$$\check{Y} = Y - \text{OLS}(Y \sim X).\text{predict}(X)$$

**Step 2:** Regress $D$ on $X$ and compute residuals—the "surprise" in $D$:

$$\check{D} = D - \text{OLS}(D \sim X).\text{predict}(X)$$

**Step 3:** Run simple line-fit on the surprises:

$$\hat{\theta} = \text{OLS}(\check{Y} \sim \check{D}) \quad \Leftrightarrow \quad \text{solve empirical normal equations } \mathbb{E}_n[(\check{Y} - \hat{\theta}\check{D})\,\check{D}] = 0$$

# Visual Intuition for FWL

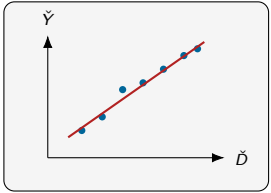
**Original Problem**

$\text{OLS}(Y \sim D, X)$

High-dimensional regression

**Same $\hat{\theta}$**

**FWL Equivalent**




A complex multivariate regression reduces to a simple line fit on residuals!

# The FWL Formula

From the simple line-fit OLS($\check{Y} \sim \check{D}$), we get:

$$\hat{\theta} = \frac{\mathbb{E}_n[\check{Y} \cdot \check{D}]}{\mathbb{E}_n[\check{D}^2]} = \frac{\mathrm{Cov}_n(\check{Y}, \check{D})}{\mathrm{Var}_n(\check{D})}$$

**Key Insight**

$\hat{\theta}$ is the **normalized correlation** between residualized outcome and residualized treatment—how much the "surprise" in $Y$ moves with the "surprise" in $D$.

*Think of it as:* When treatment is higher than expected given confounders, is the outcome also higher than expected?

# Python Pseudocode: The FWL Theorem

```
# --- Standard OLS approach ---
model_full = LinearRegression().fit(np.c_[D, X])
theta_full = model_full.coef_[0]

# --- FWL three-step procedure ---
# 1. Residualize Y: Regress Y on X
Yres = Y - LinearRegression().fit(X, Y).predict(X)
# 2. Residualize D: Regress D on X
Dres = D - LinearRegression().fit(X, D).predict(X)
# 3. Final Regression: Regress residualized Y on residualized D
theta_fwl = LinearRegression(fit_intercept=False).fit(Dres, Yres)

# --- Verification ---
print(f"Full OLS theta: {theta_full:.6f}")
print(f"FWL OLS theta:  {theta_fwl:.6f}")
```

# Poll: Interpreting OLS Coefficients

**Poll Question**

What does the coefficient $\theta$ on the first feature in OLS($Y \sim X$) represent?

A. The first coordinate of $\mathbb{E}[XX']^{-1}\mathbb{E}[XY]$

B. The correlation between $Y$ and the first feature

C. The normalized correlation between the residual of $Y$ and the residual of the first feature, after removing linear predictions from other variables

D. The average effect of changing the first feature by one unit

(Poll Everywhere)

# Live Coding: Verifying the FWL Theorem

## Exercise

**Goal:** Verify that OLS($Y \sim D, X$) gives the same coefficient as the FWL process.

**Steps:**

1. Run full OLS: $Y \sim D + X$
2. Regress $Y$ on $X$, get residuals $\check{Y}$
3. Regress $D$ on $X$, get residuals $\check{D}$
4. Regress $\check{Y}$ on $\check{D}$
5. Compare coefficients!

**Dataset:** Price elasticity simulation

Open notebook and follow along!

# The Frisch-Waugh-Lovell Theorem (in Population Limit)

**Theorem (Frisch-Waugh-Lovell) in population**

The coefficient $\theta$ on $D$ in the Best Linear Predictor $\text{BLP}(Y \sim D, X)$ is identical to the coefficient from:

**Step 1:** Calculate BLP of $Y$ using $X$ and compute *population* residuals:

$$\tilde{Y} = Y - \text{BLP}(Y \sim X).\text{predict}(X)$$

**Step 2:** Calculate BLP of $D$ using $X$ and compute *population* residuals:

$$\tilde{D} = D - \text{BLP}(D \sim X).\text{predict}(X)$$

**Step 3:** Find coefficient in BLP of $\tilde{Y}$ using $\tilde{D}$:

$$\theta = \text{coeff in BLP}(\tilde{Y} \sim \tilde{D}) \quad \Leftrightarrow \quad \text{solve normal equations } \mathbb{E}[(\tilde{Y} - \theta\tilde{D})\,\tilde{D}] = 0$$

# Answering Question 1: Interpreting $\theta$

**Question:** How can we better interpret the population coefficient $\theta$ that OLS converges to?

$$\theta = \frac{\mathbb{E}[\tilde{Y}\tilde{D}]}{\mathbb{E}[\tilde{D}^2]} = \frac{\mathrm{Cov}(\tilde{Y}, \tilde{D})}{\mathrm{Var}(\tilde{D})}$$

**Answer**

It is the slope of the line-fit between:

- The part of $Y$ not linearly predictable from $X$ (the "surprise" in outcome)
- The part of $D$ not linearly predictable from $X$ (the "surprise" in treatment)

$\theta$ captures how much the (*linearly*) unexplained variation in $Y$ moves with the (*linearly*) unexplained variation in $D$, after controlling for $X$.

*Much more intuitive than "the first coordinate of* $\mathbb{E}[XX']^{-1}\mathbb{E}[XY]$*"!*

# Answering Question 1: When is $\theta$ Uniquely Identified?

**Question:** When does the population BLP have a unique solution for $\theta$?

**Answer**

When $D$ cannot be perfectly linearly predicted from $X$—i.e., when there's some "surprise" left in treatment.

- If $D = \gamma' X$ exactly (perfect collinearity), then $\tilde{D} = 0$ and $\mathrm{Var}(\tilde{D}) = 0$
- The coefficient $\theta$ is undefined—this is the **collinearity problem**
- But this is much more benign than the overlap assumption!
- We only need $\mathrm{Var}(\check{D}) > 0$ **on average**, not pointwise overlap

# Part II: Proof of the FWL Theorem

Understanding Why It Works

# OLS as Orthogonal Decomposition

**Key Insight**

Finding OLS($Y \sim D, X$) coefficients is equivalent to finding a decomposition:

$$Y = \theta D + \beta' X + c + \epsilon$$

where $\epsilon$ is empirically orthogonal to $(1, D, X)$ (denoted as $\epsilon \perp_n (1, D, X)$), i.e.,

$$\mathbb{E}_n[\epsilon] = 0, \quad \mathbb{E}_n[\epsilon D] = 0, \quad \mathbb{E}_n[\epsilon X] = 0$$

Any (coefficients, residual) pair satisfying this orthogonality is a valid OLS($Y \sim D, X$) solution. *Think of it as: OLS extracts everything useful from the predictors. The residual has no remaining correlation.*

**Strategy:** We will show that the FWL procedure produces coefficients satisfying these normal equations.

# The Residualization Operator

Define the **residualization operator**: For any variable $V$,

$$\check{V} = V - \gamma_V' X - c_V$$

where $\gamma_V, c_V$ are chosen so that $\check{V} \perp_n (1, X)$.

*This removes the part of $V$ that's linearly predictable from $X$.*

**Key Property: Additivity**

$$\widecheck{A + B} = \check{A} + \check{B}$$

**Proof:** Write $A + B = (\gamma_A + \gamma_B)'X + (c_A + c_B) + (\check{A} + \check{B})$.
Since $\check{A} + \check{B} \perp_n (1, X)$, this is a valid decomposition. *(The surprise in $A + B$ is the sum of the surprises.)*

# Proving FWL

Start with OLS decomposition: $Y = \theta D + \beta' X + c + \epsilon$, with $\epsilon \perp_n (1, D, X)$

**Residualize both sides:**

$$\check{Y} = \theta \check{D} + \beta' \check{X} + \check{c} + \check{\epsilon}$$

**Now observe:**

- $\check{X} = 0$ ($X$ is perfectly linearly predictable from $X$)
- $\check{c} = 0$ (constant is perfectly linearly predictable)
- $\check{\epsilon} = \epsilon$ ($\epsilon \perp_n (1, X)$ means $\epsilon$ is its own residual)

**Therefore:**

$$\check{Y} = \theta \check{D} + \epsilon, \quad \text{with } \epsilon \perp_n \check{D}$$

Thus, the pair $\theta, \epsilon$ is a valid OLS($\check{Y} \sim \check{D}$) decomposition and hence the solution to the OLS($\check{Y} \sim \check{D}$) problem! □

# Part III: Variance and Standard Errors

The Insensitivity Theorem

# The Insensitivity Theorem

**Theorem (Insensitivity)**

When calculating standard errors for $\hat{\theta}$, we can ignore the estimation error in the first two OLS steps.

Asymptotically, $\hat{\theta}$ behaves as if we had the true population residuals:

$$\tilde{D} = D - \text{BLP}(D|X), \quad \tilde{Y} = Y - \text{BLP}(Y|X)$$

*You might worry: "I estimated those residuals—doesn't that add uncertainty?" Surprisingly, no!*

**Implication:** For inference, we only need to analyze a simple line-fit. Same "insensitivity" as the DR estimator!

# Deriving the Asymptotic Distribution

Starting from the population residuals:

$$\sqrt{n}(\hat{\theta} - \theta) = \sqrt{n}\left(\frac{\mathbb{E}_n[\check{D}\check{Y}]}{\mathbb{E}_n[\check{D}^2]} - \theta\right) \approx \sqrt{n}\left(\frac{\mathbb{E}_n[\tilde{D}\tilde{Y}]}{\mathbb{E}_n[\tilde{D}^2]} - \theta\right) = \sqrt{n}\frac{\mathbb{E}_n[\tilde{D}\tilde{Y}] - \theta\mathbb{E}_n[\tilde{D}^2]}{\mathbb{E}_n[\tilde{D}^2]}$$

$$= \sqrt{n}\frac{\mathbb{E}_n[\tilde{D}(\tilde{Y} - \theta\tilde{D})]}{\mathbb{E}_n[\tilde{D}^2]}$$

By LLN, denominator $\to \mathrm{Var}(\tilde{D})$. By CLT:

$$\boxed{\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} N\left(0, \frac{\mathbb{E}[\tilde{D}^2(\tilde{Y} - \theta\tilde{D})^2]}{\mathrm{Var}(\tilde{D})^2}\right)}$$

# Computing Standard Errors in Practice

The asymptotic variance can be estimated using empirical analogues:

$$\hat{\sigma}^2 = \frac{\mathbb{E}_n[\check{D}^2(\check{Y} - \hat{\theta}\check{D})^2]}{\mathrm{Var}_n(\check{D})^2}$$

Standard error: s.e. $= \hat{\sigma}/\sqrt{n}$

**Practical Note**

This is exactly the **HC0 robust standard error** that OLS packages compute! They don't calculate it this way, but it's numerically equivalent.

*Bottom line: just use robust standard errors from your favorite package—they're doing the right thing.*

# Python Pseudocode: Standard Errors via the Insensitivity Theorem

```
# Final stage residuals:
# outcome residual minus predicted from treatment residual
epsilon = Y_residuals - theta_fwl * D_residuals

# Asymptotic variance formula:
# sigma^2 = E[(Y_res - theta*D_res)^2 * D_res^2] / E[D_res^2]^2
numerator = np.mean(epsilon**2 * D_residuals**2)
denominator = np.mean(D_residuals**2)**2
sigma_sq = numerator / denominator

# Standard error = sigma / sqrt(n)
se_manual = np.sqrt(sigma_sq / len(Y))
```

# Live Coding: Computing Standard Errors

## Exercise

**Goal:** Verify that our manual SE formula matches statsmodels HC0.

**Steps:**

1. Get SE from statsmodels (HC0)
2. Compute residuals $\check{Y}, \check{D}$ from Part 1
3. Calculate $\epsilon = \check{Y} - \hat{\theta}\check{D}$
4. Apply the variance formula
5. Compare: s.e. $= \hat{\sigma}/\sqrt{n}$

Open notebook and follow along!

# Answering Question 3: Drivers of High Variance

**Question:** What are the drivers of high variance for $\hat{\theta}$?

From $\hat{\theta} = \mathrm{Cov}_n(\check{Y}, \check{D})/\mathrm{Var}_n(\check{D})$, we see two key drivers:

1. **Small denominator:** If $\mathrm{Var}(\check{D})$ is small (treatment is highly predictable from confounders), we divide by a small number $\Rightarrow$ high variance
   *Little "surprise" in treatment means little to learn from.*
2. **Large numerator noise:** If the residual $\check{Y} - \theta\check{D}$ has high variance, the estimate is noisy

**Key Insight**

High linear predictability of treatment from covariates leads to high variance—but this is much weaker than requiring overlap everywhere!

# Answering Question 4: The Analogue of Overlap

**Question:** What is the analogue of "overlap" under linearity?

**Answer**

$\mathrm{Var}(\tilde{D}) > \epsilon$ for some $\epsilon > 0$

- We need the residual treatment variance to be bounded away from zero
- This is an **average condition**, not a pointwise condition
- Even if $D$ is perfectly predictable in some regions of $X$, we're fine as long as it's unpredictable in other regions

*Compare to binary treatment overlap (*$0 < P(D = 1|X) < 1$ *everywhere)—linearity lets us extrapolate!*

# Poll 2: Variance Drivers

**Poll Question**

In OLS($Y \sim D, X$), when would you expect high variance for the coefficient on $D$?

A. When $Y$ has high variance
B. When $D$ can be well predicted linearly from $X$
C. When $X$ has many dimensions
D. When the sample size is small

(Poll Everywhere)

# Part IV: Why Insensitivity Holds

The Same Logic as Doubly Robust

# The Insensitivity Argument

Our target parameter solves: $\mathbb{E}[(\tilde{Y} - \theta\tilde{D})\tilde{D}] = 0$

Define for arbitrary first-stage coefficients $\gamma_Y, \gamma_D$:

$$M(\theta, \gamma_Y, \gamma_D) = \mathbb{E}[(Y - \gamma_Y' X - \theta(D - \gamma_D' X))(D - \gamma_D' X)]$$

**Claim**

This formula is **insensitive** to $\gamma_Y$ and $\gamma_D$ at the true values—small errors in the first stage don't affect the final answer.

*Same logic as the DR estimator!*

# Checking Insensitivity

**Partial derivative w.r.t. $\gamma_Y$:**

$$\partial_{\gamma_Y} M(\theta_0, \gamma_Y^*, \gamma_D^*) = -\mathbb{E}[X(D - \gamma_D^{*\prime} X)] = -\mathbb{E}[X\tilde{D}] = 0$$

The last equality holds because $\tilde{D} \perp X$ by construction!

**Similarly,** $\partial_{\gamma_D} M = 0$ by the orthogonality properties.

**Conclusion**

First-stage estimation errors don't affect the asymptotic distribution of $\hat{\theta}$. We can use simple standard error formulas!

# Summary: Four Questions Answered

| **Question** | **Answer** |
|---|---|
| When is $\theta$ uniquely identified? | When $D$ is not perfectly linearly predictable from $X$ (some "surprise" remains) |
| How to interpret $\theta$? | Slope of line-fit between residualized $Y$ and residualized $D$ (correlation of surprises) |
| What drives high variance? | High noise in $Y$, high linear predictability of $D$ from $X$ (little surprise) |
| Analogue of overlap? | $\mathrm{Var}(\tilde{D}) > \epsilon > 0$ (average condition, not pointwise) |

These answers come from the FWL theorem + Insensitivity theorem!

# Part V: Beyond Full Linearity

Partial Linearity and Double Machine Learning

# The Partial Linear Model

Full linearity is restrictive. What if we relax it?

**Partial Linear Model**

$$Y = \theta D + f(X) + \epsilon, \quad \mathbb{E}[\epsilon | D, X] = 0$$

- $f(X)$ can be **any function**—no linearity assumed for how confounders affect outcome
- Only the treatment effect is assumed constant: $\mathbb{E}[Y(d) - Y(0)|X] = \theta d$
- This is much more flexible while still enabling valid inference on $\theta$

*We're saying: "I don't know how confounders affect the outcome—could be complicated. But each unit of treatment has the same effect."*

# The Generalized FWL Idea

What if we replace "linearly predictable" with "predictable" in FWL?

**Generalized Double Residualization:**

1. Learn $\hat{h}(X)$ predicting $Y$ from $X$ using **any ML method**
2. Learn $\hat{p}(X)$ predicting $D$ from $X$ using **any ML method**
3. Compute residuals (surprises): $\check{Y} = Y - \hat{h}(X)$, $\check{D} = D - \hat{p}(X)$
4. Run simple line-fit: $\hat{\theta} = \text{OLS}(\check{Y} \sim \check{D})$

**Question:** Does this recover the correct $\theta$?

# Population Identification

In the population, with true conditional expectations $h_0(X) = \mathbb{E}[Y|X]$ and $p_0(X) = \mathbb{E}[D|X]$:

$$\begin{aligned}\tilde{Y} &= Y - \mathbb{E}[Y|X] = \theta D + f(X) + \epsilon - \theta\mathbb{E}[D|X] - f(X) \\ &= \theta(D - \mathbb{E}[D|X]) + \epsilon = \theta\tilde{D} + \epsilon\end{aligned}$$

Since $\mathbb{E}[\epsilon|D, X] = 0$, we have $\mathbb{E}[\epsilon|\tilde{D}] = 0$.

*The $f(X)$ terms cancel out! We don't need to know what $f$ is.*

**Conclusion**

$\theta$ is the solution to OLS($\tilde{Y} \sim \tilde{D}$) in the population!

# Does Insensitivity Still Hold?

Define for arbitrary predictive models $h, p$:

$$M(\theta, h, p) = \mathbb{E}[(Y - h(X) - \theta(D - p(X)))(D - p(X))]$$

**Check sensitivity to $h$:** Let $h_t = h_0 + t\Delta$

$$\partial_t M(\theta, h_t, p_0)|_{t=0} = -\mathbb{E}[\Delta(X)(D - p_0(X))] = -\mathbb{E}[\Delta(X)\mathbb{E}[D - p_0(X)|X]] = 0$$

**Check sensitivity to $p$:** Similarly equals 0.

**Yes!**

The formula is insensitive to the predictive models—even complex ML models! We can use powerful ML for confounders and still trust our standard errors.

# The Overfitting Problem

**Wait!**

There's a catch we saw with DR: **overfitting and double-dipping**!

If we use the same data to:

1. Train $\hat{h}$ and $\hat{p}$
2. Compute residuals
3. Fit the final regression

We may overfit and get biased estimates. *(The model "memorizes" training points, making residuals artificially small.)*

**Solution**

**Cross-fitting**, just like with DR! For each point, predict using a model trained on *other* points.

# The Double Machine Learning Algorithm

**Algorithm (Double ML with K-fold Cross-Fitting)**

1. Split data into $K$ folds
2. For each fold $k$:
   - Train $\hat{h}^{(-k)}(X)$ on out-of-fold data to predict $Y$ from $X$
   - Train $\hat{p}^{(-k)}(X)$ on out-of-fold data to predict $D$ from $X$
   - For each sample $i$ in fold $k$: $\check{Y}_i = Y_i - \hat{h}^{(-k)}(X_i)$, $\check{D}_i = D_i - \hat{p}^{(-k)}(X_i)$
3. Run simple OLS on all residuals (no intercept):

$$\hat{\theta} = \frac{\mathbb{E}_n[\check{Y}\check{D}]}{\mathbb{E}_n[\check{D}^2]}$$

# Python Pseudocode for Double ML

```
from sklearn.model_selection import KFold

def double_ml(Y, D, X, n_folds=5):
    kf = KFold(n_splits=n_folds, shuffle=True)
    Y_res, D_res = np.zeros(len(Y)), np.zeros(len(Y))

    for train, test in kf.split(X):
        # Train on out-of-fold, predict for in-fold;
        # AutoML is placeholder for any ML method
        h_model = AutoML().fit(X[train], Y[train])
        Y_res[test] = Y[test] - h_model.predict(X[test])

        p_model = AutoML().fit(X[train], D[train])
        D_res[test] = D[test] - p_model.predict(X[test])

    # Final line-fit on surprises
    theta_hat = np.mean(Y_res * D_res) / np.mean(D_res**2)
    return theta_hat
```

# Asymptotic Normality of Double ML

**Theorem**

Under regularity conditions, the cross-fitted DML estimator is asymptotically normal:

$$\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} N\left(0, \frac{\mathbb{E}[(\tilde{Y} - \theta\tilde{D})^2\tilde{D}^2]}{\mathrm{Var}(\tilde{D})^2}\right)$$

**Required conditions** (the ML models need to be "good enough"):

1. Product rate: $\sqrt{n} \cdot \mathrm{RMSE}(\hat{h}) \cdot \mathrm{RMSE}(\hat{p}) \to 0$
2. Treatment model rate: $n^{1/4} \cdot \mathrm{RMSE}(\hat{p}) \to 0$
3. Consistency: $\mathrm{RMSE}(\hat{h}) \to 0$

*We saw already how Random forests, Lasso, Neural Nets typically satisfy these conditions under small "effective dimension" assumptions.*

# Standard Errors for Double ML

Estimate the asymptotic variance using empirical analogues:

$$\hat{\sigma}^2 = \frac{\mathbb{E}_n[(\check{Y} - \hat{\theta}\check{D})^2 \check{D}^2]}{\mathbb{E}_n[\check{D}^2]^2}$$

Standard error: s.e. $= \hat{\sigma}/\sqrt{n}$

```
def double_ml_with_se(Y, D, X, n_folds=5):
    theta_hat, Y_res, D_res = double_ml(Y, D, X, n_folds)

    # Compute standard error
    epsilon = Y_res - theta_hat * D_res
    sigma_sq = np.mean(epsilon**2 * D_res**2) / np.mean(D_res**2)**2
    se = np.sqrt(sigma_sq / len(Y))

    return theta_hat, se
```

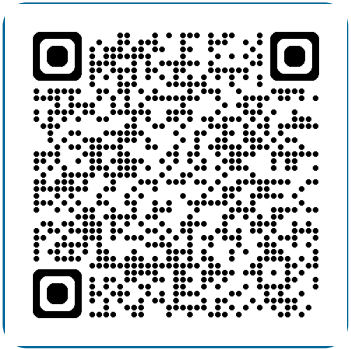
# Live Coding: Implementing Double ML

## Exercise

**Goal:** Implement DML from scratch with cross-fitting.

**Steps:**

1. Implement cross-fitting loop
2. Use FLAML AutoML to find best models
3. Compute residuals for each fold
4. Run final line-fit on residuals
5. Compute standard errors
6. Compare with true elasticity!



Open notebook and follow along!

**Key insight:** DML recovers the true effect even when $f(X)$ is nonlinear!

# The Overlap Analogue in Partial Linearity

**Key Insight**

The overlap condition becomes $\mathrm{Var}(\tilde{D}) > \epsilon$

- $\tilde{D} = D - \mathbb{E}[D|X]$ is the unpredictable part of treatment
- We need this to have positive variance **on average**
- Much weaker than pointwise overlap!

**For binary treatments:** $\mathrm{Var}(\tilde{D}) = \mathbb{E}[p(X)(1 - p(X))]$

- We need propensity bounded away from 0 and 1 **on average**
- Can extrapolate across covariate regions under partial linearity

# Part VI: Interpretation with Mis-Specification

What Does DML Estimate if CEF is not Partially Linear?

# Treatment Effect Heterogeneity: Binary Case

What if the true model has heterogeneous effects?

$$\mathbb{E}[Y|D,X] = \tau(X) \cdot D + f(X)$$

where $\tau(X)$ varies with $X$. We can still run the DML procedure. Irrespective of partial linearity, DML will converge to the quantity:

$$\theta_{\text{DML}} = \frac{\mathbb{E}[\tilde{D}\tilde{Y}]}{\text{Var}(\tilde{D})}, \quad \text{with } \tilde{Y} = Y - \mathbb{E}[Y \mid X] \text{ and } \tilde{D} = D - \mathbb{E}[D \mid X]$$

**Overlap-Weighted Average Treatment Effect**

DML estimates a weighted average of $\tau(X)$, with weights $\propto \text{Var}(D|X)$—more weight where treatment is more "random" (less predictable), i.e.,

$$\theta_{\text{DML}} = \frac{\mathbb{E}[\tilde{D}\tilde{Y}]}{\text{Var}(\tilde{D})} = \frac{\mathbb{E}[\tau(X)\text{Var}(D|X)]}{\mathbb{E}[\text{Var}(D|X)]}$$

# Why Use DML Even for Binary Treatments?

Even though DR works for binary treatments, DML has advantages:

1. **Lower variance:** Weights by $\mathrm{Var}(D|X) = p(X)(1 - p(X))$, downweighting extreme propensities
2. **Natural interpretation:** Overlap-weighted ATE
3. **Robustness to overlap violations:** Works even with some regions of poor overlap

**Trade-off**

Has lower variance and does not require overlap, but requires partial linearity assumption (constant treatment effect). If effects are heterogeneous, DML gives a weighted average, not the population ATE.

# Python Code for Binary Treatment DML

```
from sklearn.linear_model import LogisticRegression

def double_ml_binary(Y, D, X, n_folds=5):
    kf = KFold(n_splits=n_folds, shuffle=True)
    Y_res, D_res = np.zeros(len(Y)), np.zeros(len(Y))

    for train, test in kf.split(X):
        # Outcome model
        h_model = AutoML().fit(X[train], Y[train])
        Y_res[test] = Y[test] - h_model.predict(X[test])

        # Propensity model (classifier for binary D)
        p_model = AutoMLClassifier().fit(X[train], D[train])
        D_res[test] = D[test] - p_model.predict_proba(X[test])[:,1]

    theta_hat = np.mean(Y_res * D_res) / np.mean(D_res**2)
    return theta_hat
```

# Continuous Treatments: Weighted Average Derivatives

For continuous treatments **without** partial linearity, let $g(d, X) = \mathbb{E}[Y|D = d, X]$.

Even without assuming $g(d, X) = \theta d + f(X)$, DML estimates a **weighted average marginal effect**:

$$\theta_{\text{DML}} = \mathbb{E}\left[ w(D, X) \cdot \frac{\partial}{\partial d} g(d, X)\Big|_{d=D} \right]$$

for some weight function $w(D, X)$.

*Think of it as: "On average, how much does outcome change when we nudge treatment?"*

The weights are complex in general, but DML still has a causal interpretation as a weighted average derivative!

# Special Case: Gaussian Treatment

When $D|X \sim N(p(X), \sigma^2(X))$, the story simplifies beautifully. The DML numerator is:

$$\mathbb{E}[\tilde{Y} \cdot \tilde{D}] = \mathbb{E}[Y \cdot \tilde{D}] = \mathbb{E}[g(D, X) \cdot (D - p(X))]$$

By **Stein's Lemma** (a result for Gaussian random variables):

$$\mathbb{E}[g(D, X) \cdot (D - p(X))] = \mathbb{E}\left[\sigma^2(X) \cdot \frac{\partial}{\partial d} g(d, X)\Big|_{d=D}\right]$$

**Variance-Weighted Average Marginal Effect**

$$\theta_{\text{DML}} = \frac{\mathbb{E}\left[\sigma^2(X) \cdot \partial_d g(D, X)\right]}{\mathbb{E}[\sigma^2(X)]}$$

# Interpretation of the Gaussian Case

**Key Result**

Under Gaussian treatment, DML estimates the **variance-weighted average marginal effect**:

$$\theta_{\text{DML}} = \frac{\mathbb{E}\left[\sigma^2(X) \cdot \partial_d g(D, X)\right]}{\mathbb{E}[\sigma^2(X)]}$$

**Intuition:**

- $\partial_d g(D, X)$ is the marginal effect at $(D, X)$
- $\sigma^2(X)$ is the conditional variance of treatment given $X$
- We weight by variance: more weight where treatment is more "experimental"

*Even without partial linearity, DML has a clean causal interpretation!*

# Part VII: Industry Applications

Double ML in Practice

# DML at Amazon

Amazon paper claims usage of DML in their experimentation platform for causal inference at scale.

**Paper:** "Double Machine Learning at Scale to Predict Causal Impact of Customer Actions" (Amazon Science)

**Key applications:**

- Estimating effect of customer actions on purchases
- Price elasticity estimation
- Marketing attribution

**Why DML?** Handles high-dimensional covariates, flexible ML models, valid inference, continuous treatments, lower variance that DR, less sensitive to overlap!

# Software Implementations

**EconML (Microsoft):**

```
from econml.dml import LinearDML

est = LinearDML(model_y=AutoML(), model_t=AutoML())
est.fit(Y, D, X=None, W=X)
est.summary()
```

**DoubleML (Python package):**

```
from doubleml import DoubleMLPLR
from sklearn.ensemble import RandomForestRegressor

ml_g = RandomForestRegressor()
ml_m = RandomForestRegressor()
dml = DoubleMLPLR(data, ml_g, ml_m)
dml.fit()
dml.summary
```

# Live Coding: Comparing with Libraries

## Exercise

**Goal:** Compare our implementation with EconML and DoubleML.

**Steps:**

1. Run EconML LinearDML
2. Run DoubleML DoubleMLPLR
3. Compare all estimates:
   - OLS (linear controls)
   - Our DML implementation
   - EconML
   - DoubleML
4. Interpret as price elasticity!

Open notebook and follow along!

# Summary: Four Key Takeaways

1. **The Frisch-Waugh-Lovell Theorem:** The coefficient $\theta$ in OLS$(Y \sim D, X)$ equals the slope from regressing the "surprise" in $Y$ on the "surprise" in $D$—the normalized correlation of residuals.
2. **Insensitivity to First-Stage Errors:** When computing standard errors for $\theta$, we can ignore estimation error in the first-stage regressions—another "insensitive formula" like the DR estimator.
3. **Partial Linearity Enables Flexibility:** Assuming $\mathbb{E}[Y|D, X] = \theta D + f(X)$ allows ML for $f(X)$ while still getting valid inference on $\theta$.
4. **Double ML and Weighted Average Effects:** DML estimates variance-weighted average treatment effects (or marginal effects under Gaussian treatments via Stein's lemma)—widely used in industry.

# Next Time

- Beyond partial linearity?
- General principle and framework of "insensitivity" for causal ML estimation problems

# References


- Chernozhukov, V., et al. (2018). Double/debiased machine learning for treatment and structural parameters. *The Econometrics Journal*, 21(1), C1-C68.
- Frisch, R., & Waugh, F. V. (1933). Partial time regressions as compared with individual trends. *Econometrica*, 1(4), 387-401.
- Lovell, M. C. (1963). Seasonal Adjustment of Economic Time Series and Multiple Regression Analysis. *Journal of the American Statistical Association*, 58(304), 993–1010.
- Amazon Science. Double machine learning at scale to predict causal impact of customer actions.
- EconML Documentation: `https://econml.azurewebsites.net/`
- DoubleML Documentation: `https://docs.doubleml.org/`